Detection of Subpixel Targets on Hyperspectral Remote Sensing Imagery Based on Background Endmember Extraction

Xiaorui Song, Ling Zou, and Lingda Wu

***Abstract*— The low spatial resolution associated with imaging spectrometers has caused subpixel target detection to become a special problem in hyperspectral image (HSI) processing that poses considerable challenges. In subpixel target detection, the size of the target is smaller than that of a pixel, making the spatial information of the target almost useless so that a detection algorithm must rely on the spectral information of the image. To address this problem, this article proposes a subpixel target detection algorithm for hyperspectral remote sensing imagery based on background endmember extraction. First, we propose a background endmember extraction algorithm based on robust nonnegative dictionary learning to obtain the background endmember spectrum of the image. Next, we construct a hyperspectral subpixel target detector based on pixel reconstruction (HSPRD) to perform pixel-by-pixel target detection on the image to be tested using the background endmember spectral matrix and the spectra of known ground targets. Finally, the subpixel target detection results are obtained. The experimental results show that, compared with other existing subpixel target detection methods, the algorithm proposed here can provide the optimum target detection results for both synthetic and real-world data sets.**

***Index Terms*— Background endmember extraction, hyperspectral image (HSI), robust nonnegative dictionary learning, subpixel, target detection.**

I. Introduction

As the technology of hyperspectral imaging continues to develop, newer imaging spectrometers can directly collect hundreds of spectral channels from the Earth's surface. A final acquired hyperspectral remote sensing image can accurately represent the chemical composition of different ground features. Hence, while providing spatial information, hyperspectral remote sensing images can simultaneously provide the users with diagnostic spectral information that differentiates features other than simply providing spatial data [1]. This process is very valuable in the field of ground object detection. Given the spectral complexity and diversity of natural features and the low spatial resolution of the spectral sensors, a small-sized ground target may occupy part of the pixel area and form a mixed pixel with the background of a ground object. In such cases, the spectral features exhibited by the mixed pixel will show the absorption characteristics of multiple endmembers, and their shapes may differ from that of the spectral features of ground targets. Therefore, when a ground target cannot fill up an entire pixel, the corresponding target detection problem becomes a subpixel target detection problem because only one endmember of the mixed pixel is of interest. This type of problem can no longer be solved via the spatial information and a simple match of the spectrum.

The key to target detection is the separation of the target from the background. The subpixel target detection methods available today can be classified based on the structure of the background model. In one class of methods, known as nonstructural background detection methods, statistical distribution is used to describe the variation in the background. This class mainly includes the matched filter method [2], the adaptive coherence estimator method [3], the adaptive matched filter method [4], and two methods developed on this basis: the subpixel spectral matched filter (SPSMF) method and the segmented adaptive coherence estimator (SACE) method [5]. In another class, known as structured background detection methods, subspace models are used to describe the variation in the background. This class mainly includes the orthogonal subspace projection method [6], the adaptive subspace detector method [7], and the kernel orthogonal subspace projection detector method [8], all of which have been developed on this basis. Yet, another class of detection methods, based on a linear combination of models, is also currently popular. Its mathematical model is similar to that of the subspace model, but the endmember spectra that constitute the observational data have a more defined physical meaning and represent the composition of the ground object in the image. The main methods of this class include the hybrid subpixel detector of Broadwater *et al.* [9], which combines a fully constrained least-squares algorithm, the adaptive matched subspace detector (AMSD), adaptive coherence estimator detection methods, and the hybrid selective endmembers' unstructured detector method [10] proposed by Du *et al.* and the hybrid selective endmember structures' detector method developed on this basis.

Along with the broad application of the sparse representation theory in image processing, some researchers have introduced the sparse representation theory to

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the hyperspectral subpixel target detection problem. By establishing a background model using the sparseness of the hyperspectral data, the results of target detection may be obtained by combining it with a binary detection algorithm. Representative algorithms include the combined sparse and collaborative representation (CSCR) algorithm [11] for supervised target detection, the background joint sparse representation detection method [12] for subpixel target detection, and the subpixel target detection method based on dictionary reconstruction [13]. Although these methods reconstruct the background using the sparseness of the hyperspectral data, their loss functions all take the form of an l_2-norm, which are sensitive to the few ground object targets regarded as anomalies in background reconstruction and, to a certain degree, are affected the accuracy of the sparse reconstruction. Moreover, in target detection after background reconstruction, the detection operators employed do not have sufficient sensitivity to detect small targets that only occupy a small fraction of the pixels.

In order to improve the accuracy of the background reconstruction of a hyperspectral image (HSI), we used the method of spectral dictionary learning to perform hybrid pixel decomposition on the detected image and to extract the background endmember while obtaining the corresponding abundance coefficient. On this basis, we introduce a new form of detection operator to make the targets of low occupancy and more sensitive and to obtain better detection results.

The two main contributions of this article are as follows. First, an endmember extraction algorithm based on robust nonnegative dictionary learning is proposed. The algorithm improves the robustness to the outlier points and the accuracy for extracting the background endmember spectrum. Second, a new form of target spectral detection operator is introduced to improve the detection accuracy for small targets.

The contents of this article are arranged as follows. Section I provides an introduction. Section II gives an analysis of the subpixel target detection model. Section III proposes a background endmember extraction method based on robust nonnegative dictionary learning. Section IV proposes a subpixel target detection operator based on pixel reconstruction. Section V proposes an experimental verification of the proposed subpixel target detection algorithm. Section VI proposes a summary.

II. Subpixel Target Detection Model

According to the replacement model stated in [14], any pixel $\boldsymbol{x}$ in the observed image can be expressed as follows:

$$\boldsymbol{x} = m\boldsymbol{s} + (1-m)\boldsymbol{v} + \boldsymbol{n}, \quad 0 \le m \le 1 \tag{1}$$

where $\boldsymbol{s}$ represents the target spectral signature, $\boldsymbol{v}$ represents the background spectral signature, m is denoted as the target fill factor, and $\boldsymbol{n}$ represents the additive noise originated from the atmospheric or the imaging spectrometer. Obviously, in practice, m is always unknown. When the target fill factor is very small or very large ($0 < m \ll 1$ or $m \approx 1$), a good approximation of the replacement model is the widely used additive model [14]. Since the assumptions in the additive model are not valid for real HSI data, in this article, we use the replacement model to improve modeling accuracy.

Based on the binary hypothesis of the test model, (1) can be rewritten as

$$\begin{aligned} H_0 &: \boldsymbol{x} = \boldsymbol{v} + \boldsymbol{n} \quad \text{(Target absent)} \\ H_1 &: \boldsymbol{x} = m\boldsymbol{s} + (1-m)\boldsymbol{v} + \boldsymbol{n} \quad \text{(Target present)} \end{aligned} \tag{2}$$

where H_0 and H_1 are the binary opposite hypotheses.

In this article, it was assumed that the observed spectrum of the background pixel is a linear combination of the background endmember spectra, and the observed spectrum of the target pixel is a linear combination of the target and background spectra. Thus, we set

$$\begin{aligned} \boldsymbol{v} &= \boldsymbol{D}_\mathrm{b}\boldsymbol{a}_\mathrm{b} \quad (\text{Under } H_0,\ m = 0) \\ m\boldsymbol{s} &= \boldsymbol{D}_\mathrm{t}\boldsymbol{a}_\mathrm{t} \quad (\text{Under } H_1,\ 0 < m \le 1) \\ (1-m)\boldsymbol{v} &= \boldsymbol{D}_\mathrm{b}\boldsymbol{a'}_\mathrm{b} \quad (\text{Under } H_1,\ 0 < m \le 1) \end{aligned} \tag{3}$$

where $\boldsymbol{D}_\mathrm{b} = [\boldsymbol{d}_{\mathrm{b}1}, \boldsymbol{d}_{\mathrm{b}2}, \ldots, \boldsymbol{d}_{\mathrm{b}k}] \in \mathbf{R}^{L\times k}$ is the background spectral dictionary, L is the number of spectral bands contained in each pixel, k is the number of background spectrum dictionary atoms, that is, the number of background endmember spectra, $\boldsymbol{d}_\mathrm{b}$ is the background dictionary spectral vector, $\boldsymbol{D}_\mathrm{t} = [\boldsymbol{d}_{\mathrm{t}1}, \boldsymbol{d}_{\mathrm{t}2}, \ldots, \boldsymbol{d}_{\mathrm{t}q}] \in \mathbf{R}^{L\times q}$ is the target spectrum dictionary, q is the number of dictionary atoms of the target spectrum, that is, the number of target endmember spectra, $\boldsymbol{d}_\mathrm{t}$ is the target dictionary spectral vector, $\boldsymbol{a}_\mathrm{b} \in \mathbf{R}^{k\times 1}$ is the abundance coefficient vector corresponding to the background spectral dictionary under H_0, $\boldsymbol{a'}_\mathrm{b} \in \mathbf{R}^{k\times 1}$ is the abundance coefficient vector corresponding to the background spectral dictionary under H_1, and $\boldsymbol{a}_\mathrm{t} \in \mathbf{R}^{q\times 1}$ is the abundance coefficient matrix vector corresponding to the target spectrum dictionary.

Therefore, (2) can be expressed as

$$\begin{aligned} H_0 &: \boldsymbol{x} = \boldsymbol{D}_\mathrm{b}\boldsymbol{a}_\mathrm{b} + \boldsymbol{n} \quad \text{(Target absent)} \\ H_1 &: \boldsymbol{x} = \boldsymbol{D}_\mathrm{b}\boldsymbol{a'}_\mathrm{b} + \boldsymbol{D}_\mathrm{t}\boldsymbol{a}_\mathrm{t} + \boldsymbol{n} \quad \text{(Target present).} \end{aligned} \tag{4}$$

Because the abundance in the HSI should satisfy sum-to-one constraint (ASC), we have

$$\begin{aligned} \sum_{i=1}^{k} a_{(i)\mathrm{b}} &= 1 \quad \text{(Target absent)} \\ \sum_{i=1}^{k} a'_{(i)\mathrm{b}} + \sum_{j=1}^{q} a_{(j)t} &= 1 \quad \text{(Target present).} \end{aligned} \tag{5}$$

When the target spectrum exist, because of the target signal "replaces" part of the signal of background, we have $\sum_{i=1}^{k} a'_{(i)\mathrm{b}} < \sum_{i=1}^{k} a_{(i)\mathrm{b}}$. In addition, by analyzing the physical significance of (2) and (5), the mathematical relationship between the abundance coefficient and the target fill factor can be obtained as follows:

$$\begin{aligned} \sum_{j=1}^{q} a_{(j)t} &= m \\ \sum_{i=1}^{k} a'_{(i)\mathrm{b}} &= 1 - m. \end{aligned} \tag{6}$$

According to the subpixel target detection model described in (4), when the target feature spectrum is known, the first step

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in target detection is to acquire the background spectrum dictionary, that is, to solve $\boldsymbol{D}_{\mathrm{b}}$. The essence of this problem is the unsupervised endmember extraction of the HSI. Because the target spectrum $\boldsymbol{D}_{\mathrm{t}}$ exists in very few pixels of the image, it can be identified as outlier points when solving $\boldsymbol{D}_{\mathrm{b}}$. Therefore, to improve the accuracy of background endmember extraction and enhance the robustness of the outliers, we introduce a robust loss function in the solution of $\boldsymbol{D}_{\mathrm{b}}$.

III. Background Endmember Extraction Based on Robust Nonnegative Dictionary Learning

According to dictionary learning and the theory of sparse representation, the background endmember matrix $\boldsymbol{D}_{\mathrm{b}}$ in (4) may be regarded as a dictionary matrix obtained by learning from the image to be detected, wherein each dictionary atom corresponds to a background endmember spectrum that constitutes the mixed pixel. The corresponding abundance coefficient vector $\boldsymbol{a}_{\mathrm{b}}$ is treated as a sparse coding. Therefore, the background endmember matrix may be obtained by a dictionary learning method. Subsequently, the abundance matrix is solved based on the obtained background endmember.

Based on this idea, an HSI unmixing method based on online dictionary learning is proposed in [15]. In this method, the standard dictionary learning format is applied, that is, the spectral dictionary is obtained by optimizing the objective function in the least absolute shrinkage and selection operator (LASSO) format, such as

$$\min_{\boldsymbol{a}_{\mathrm{b}i},\boldsymbol{D}_{\mathrm{b}}} \sum_i \frac{1}{2}\|\boldsymbol{x}_i - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}_{\mathrm{b}i}\|_2^2 + \lambda\|\boldsymbol{a}_{\mathrm{b}i}\|_1 \tag{7}$$

where $\lambda > 0$ is a regularization parameter, $\boldsymbol{X}_i$ is the ith pixel spectral vector in the HSI matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n] \in \mathbf{R}^{L\times n}$, and n is the number of pixels contained in the hyperspectral data to be detected.

In hyperspectral subpixel target detection, a target pixel in the image data may be regarded as an outlier in the background endmember extraction process, and the l_2-norm loss function applied in (7) is sensitive to outliers and noise. This can easily lead to large deviations and poor accuracy in background endmember spectral extraction. For this reason, in this article, we replace the l_2-norm loss function with a more robust l_1-norm loss function. In addition, in the actual problem, the original HSI pixel $\boldsymbol{x}$, the background spectral dictionary $\boldsymbol{D}_{\mathrm{b}}$, and the corresponding abundance coefficient $\boldsymbol{a}_{\mathrm{b}}$ are all nonnegative.

In summary, in the unsupervised endmember extraction method proposed in this article, the dictionary matrix is obtained by optimizing the following objective function or the background spectral endmember matrix:

$$\begin{aligned}&\min_{\boldsymbol{a}_{\mathrm{b}i},\boldsymbol{D}_{\mathrm{b}}} \sum_i \|\boldsymbol{x}_{\mathrm{b}i} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}_{\mathrm{b}i}\|_1 + \lambda\|\boldsymbol{a}_{\mathrm{b}i}\|_1 + \beta\|\boldsymbol{D}_{\mathrm{b}}\|_{\mathrm{F}}^2\\ &\text{s.t. } \boldsymbol{D}_{\mathrm{b}} \geq 0, \quad \boldsymbol{a}_{\mathrm{b}i} \geq 0\end{aligned} \tag{8}$$

where the smoothing term $\|\boldsymbol{D}_{\mathrm{b}}\|_{\mathrm{F}}^2$ is used to avoid trivial solutions.

Writing (8) in the matrix form, we have

$$\begin{aligned}&\min_{\boldsymbol{A}_{\mathrm{b}},\boldsymbol{D}_{\mathrm{b}}} \|\boldsymbol{X} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}}\|_1 + \lambda\|\boldsymbol{A}_{\mathrm{b}}\|_1 + \beta\|\boldsymbol{D}_{\mathrm{b}}\|_{\mathrm{F}}^2\\ &\text{s.t. } \boldsymbol{D}_{\mathrm{b}} \geq 0, \quad \boldsymbol{A}_{\mathrm{b}} \geq 0\end{aligned} \tag{9}$$

where $\boldsymbol{A}_{\mathrm{b}} = [\boldsymbol{a}_{\mathrm{b}1}, \boldsymbol{a}_{\mathrm{b}2}, \ldots, \boldsymbol{a}_{\mathrm{b}n}] \in \mathbf{R}^{k\times n}$ is the abundance coefficient matrix, and $\|\boldsymbol{A}_{\mathrm{b}}\|_1 = \sum_{ki} |a_{ki}^{\mathrm{b}}|$.

Equation (9) is the objective function to be optimized in solving the dictionary matrix or the background spectral matrix.

Equation (9) is optimized to be nonconvex, but, when one of the variables is known, we can find the global optimal solution of the other variable. Given that two nonsmooth l_1 terms exist in (9), it is somewhat challenging to solve. However, because the l_1 term in (9) appears together with the nonnegative constraint, it is feasible to solve (9). The l_1 terms related to sparse coding can be written as

$$\|\boldsymbol{A}_{\mathrm{b}}\|_1 = Tr(\boldsymbol{E}\boldsymbol{A}_{\mathrm{b}}) \tag{10}$$

where all the elements of $\boldsymbol{E} \in \mathbf{R}^{k\times n}$ are equal to 1.

Therefore, in order to solve the optimization problem of (9), the construction matrix $\boldsymbol{W}$ satisfies

$$w_{ij} = \left((\boldsymbol{X} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}})_{ij}^2 + \delta^2\right)^{-1/2} \tag{11}$$

where δ is a small number that prevents w_{ij} from overflowing in the case of $(\boldsymbol{X} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}})_{ji}^2 \simeq 0$. In the experiment here, the accuracy of δ was set to that of the machine (simulation software). When $\delta \to 0$, we have

$$\left((\boldsymbol{X} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}})_{ij}^2 + \delta^2\right)^{1/2} \to |(\boldsymbol{X} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}})_{ij}|. \tag{12}$$

For nonzero δ, the final objective function for solving the background endmember spectral matrix is

$$\begin{aligned}\min_{\boldsymbol{D}_{\mathrm{b}}\geq 0,\boldsymbol{A}_{\mathrm{b}}\geq 0}\Bigg(&\sum_i^n\sum_j^L \left((\boldsymbol{X}_{\mathrm{b}} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}})_{ji}^2 + \delta^2\right)^{1/2}\\ &+\lambda\sum_i^n\sum_j^L |\boldsymbol{A}_{ij}^{\mathrm{b}}| + \beta\|\boldsymbol{D}_{\mathrm{b}}\|_{\mathrm{F}}^2\Bigg).\end{aligned} \tag{13}$$

According to the theory in [16] related to nonnegative dictionary learning, when the variables $\boldsymbol{A}_{\mathrm{b}}$ and $\boldsymbol{D}_{\mathrm{b}}$ are fixed separately, the multiplicative update rule for the other variable can be obtained as follows:

$$d_{jk}^{\mathrm{b}} \leftarrow d_{jk}^{\mathrm{b}} \frac{\left[\boldsymbol{X} \odot \boldsymbol{W}\boldsymbol{A}_{\mathrm{b}}{}^T\right]_{jk}}{\left[(\boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}}) \odot \boldsymbol{W}\boldsymbol{A}_{\mathrm{b}}{}^T + 2\beta\boldsymbol{D}_{\mathrm{b}}\right]_{jk}} \tag{14}$$

$$a_{ki}^{\mathrm{b}} \leftarrow a_{ki}^{\mathrm{b}} \frac{\left[\boldsymbol{D}_{\mathrm{b}}{}^{\mathrm{T}}\boldsymbol{X} \odot \boldsymbol{W}\right]_{ki}}{\left[\boldsymbol{D}_{\mathrm{b}}{}^{\mathrm{T}}(\boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}}) \odot \boldsymbol{W} + \lambda\boldsymbol{E}\right]_{ki}} \tag{15}$$

where $\odot$ is the Hadamard product, that is, the product of the elements of the two matrices. Here, we assume that the Hadamard product has a higher operator precedence, i.e., $\boldsymbol{A}\boldsymbol{B} \odot \boldsymbol{C}\boldsymbol{D} = \boldsymbol{A}(\boldsymbol{B} \odot \boldsymbol{C})\boldsymbol{D}$.

The background endmember matrix $\boldsymbol{D}_{\mathrm{b}}$ can finally be obtained by solving (14) and (15).

To summarize, the background endmember extraction pseudocode based on the robust nonnegative dictionary learning can be obtained as follows.

After obtaining the background endmember matrix through the abovementioned process, the corresponding abundance coefficient matrix can then be solved. Given the distribution characteristics of the features in the actual scene, although the entire HSI scene contains many endmembers of the features, each mixed pixel usually consists of only a few

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Algorithm 1 Background Endmember Extraction Based on Robust Nonnegative Dictionary Learning

Input: Original HSI $\boldsymbol{X} \in \mathbf{R}^{L\times n}$, iteration number $T \in \mathbf{N}$, expected error tol

Output: Background endmember matrix $\boldsymbol{D}_{\mathrm{b}} \in \mathbf{R}^{L\times k}$

1 Pre-processing: Estimate number of endmembers k in the image using HySime algorithm [17]

2 Initialization: Obtain initial background endmember matrix $\boldsymbol{D}_{\mathrm{b}0} \in \mathbf{R}^{L\times k}$ using VCA algorithm. Obtain initial abundance coefficient matrix $\boldsymbol{A}_{\mathrm{b}0} \in \mathbf{R}^{k\times n}$ using nonnegative least-squares inverse solution: $\boldsymbol{A}_{\mathrm{b}0} = \arg\min_{\boldsymbol{A}_{\mathrm{b}}\geq 0}\|\boldsymbol{X} - \boldsymbol{D}_{\mathrm{b}0}\boldsymbol{A}_{\mathrm{b}0}\|_{\mathrm{F}}^2$

3 begin

4 $w_{ij} = ((\boldsymbol{X} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}})_{ij}^2 + \delta^2)^{-1/2}$

5 for $t = 1$ to T do

/* Update endmember matrix (dictionary) */

6 $d_{jk}^{t+1} \leftarrow d_{jk}^{t}\frac{[\boldsymbol{X}\odot\boldsymbol{W}\boldsymbol{A}_{\mathrm{b}}^{\mathrm{T}}]_{jk}}{[(\boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}})\odot\boldsymbol{W}\boldsymbol{A}_{\mathrm{b}}^{\mathrm{T}}+2\beta\boldsymbol{D}_{\mathrm{b}}]_{jk}}$

/* Update abundance coefficient matrix (encoding) */

7 $a_{ki}^{t+1} \leftarrow a_{ki}^{t}\frac{[\boldsymbol{D}_{\mathrm{b}}^{\mathrm{T}}\boldsymbol{X}\odot\boldsymbol{W}]_{ki}}{[\boldsymbol{D}_{\mathrm{b}}^{\mathrm{T}}(\boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}})\odot\boldsymbol{W}+\lambda\boldsymbol{E}]_{ki}}$

/* Reconstruct the HSI */

8 $\boldsymbol{X}^{t+1} = \boldsymbol{D}_{\mathrm{b}}^{t+1}\boldsymbol{A}_{\mathrm{b}}^{t+1}$

/* Calculate reconstruction error */

9 $\varepsilon = \|\boldsymbol{X}^{t+1} - \boldsymbol{X}^{t}\|_{\mathrm{F}}/\|\boldsymbol{X}^{t}\|_{\mathrm{F}}$

10 until $\varepsilon <$ tol

11 end for

12 end

(2–3) endmembers. Therefore, the corresponding abundance coefficient vector $\boldsymbol{a}_{\mathrm{b}}$ of each mixed pixel may be regarded as sparse, and the abundance coefficient matrix $\boldsymbol{A}_{\mathrm{b}}$ may be solved using the sparse unmixing method.

In addition, considering the relationship between each mixed pixel and its neighboring pixels, the abundance value belonging to the same endmember should have a smooth transition, so the total variation (TV) terms are introduced in the unmixing process. Then, the abundance coefficient matrix can be solved by the following minimization method:

$$\begin{aligned}&\min_{\boldsymbol{A}_{\mathrm{b}}}\frac{1}{2}\|\boldsymbol{D}_{\mathrm{b}}\boldsymbol{A}_{\mathrm{b}} - \boldsymbol{X}_{\mathrm{b}}\|_{\mathrm{F}}^2 + \lambda_c\|\boldsymbol{A}_{\mathrm{b}}\|_{1,1} + \lambda_{\mathrm{TV}}(\boldsymbol{A}_{\mathrm{b}})\\&\text{s.t. } \boldsymbol{A}_{\mathrm{b}} \geq \mathbf{0}\end{aligned} \tag{16}$$

where

$$\mathrm{TV}(\boldsymbol{A}_{\mathrm{b}}) \equiv \sum_{\{u,v\}\in\varphi}\|\boldsymbol{a}_{\mathrm{b}u} - \boldsymbol{a}_{\mathrm{b}v}\|_1 \tag{17}$$

is a vector extension of the anisotropic TV that ensures a smooth change of the abundance coefficient of the same kind of endmember in the adjacent domain.

The minimization problem (16) may be regarded as a constrained basis pursuit denoising method that takes neighboring information into consideration. It can be solved by the method of TV, i.e., the variable separation-enhanced Lagrangian unmixing. The abundance coefficient matrix $\boldsymbol{A}_{\mathrm{b}}$ can finally be obtained.

IV. Hyperspectral Subpixel Target Detector Based on Pixel Reconstruction

Based on the background endmember matrix $\boldsymbol{D}_{\mathrm{b}}$ obtained in Section III for the HSI to be detected and the known target feature spectrum $\boldsymbol{D}_{\mathrm{t}}$, the endmember matrix, including all the feature spectra, may be expressed as $\boldsymbol{D} = [\boldsymbol{D}_{\mathrm{b}} \;\; \boldsymbol{D}_{\mathrm{t}}] \in \mathbf{R}^{L\times(k+q)}$. For endmember matrix $\boldsymbol{D}$, the corresponding abundance coefficient matrix $\boldsymbol{A} = [\boldsymbol{A}_{\mathrm{b}} \;\; \boldsymbol{A}_{\mathrm{t}}] \in \mathbf{R}^{(k+q)\times n}$ can be obtained by the inverse solution of the nonnegative least-squares method.

Since each pixel in the hyperspectral data to be detected satisfies the linear mixed model, the background pixel can be reconstructed as

$$\boldsymbol{x} \approx \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}_{\mathrm{b}} \tag{18}$$

using the endmember matrix $\boldsymbol{D}_{\mathrm{b}}$ and the abundance coefficient vector $\boldsymbol{a}_{\mathrm{b}}$.

For a pixel containing the target feature spectrum, the reconstruction using the endmember matrix $\boldsymbol{D}$ that is included both the target and background feature spectra is more accurate and has smaller error. Thus, we have

$$\boldsymbol{x} \approx \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}'_{\mathrm{b}} + \boldsymbol{D}_{\mathrm{t}}\boldsymbol{a}_{\mathrm{t}} = \begin{bmatrix}\boldsymbol{D}_{\mathrm{b}} & \boldsymbol{D}_{\mathrm{t}}\end{bmatrix}\begin{bmatrix}\boldsymbol{a}'_{\mathrm{b}}\\ \boldsymbol{a}_{\mathrm{t}}\end{bmatrix} = \boldsymbol{D}\boldsymbol{a} \tag{19}$$

where $\boldsymbol{a} = [\boldsymbol{a}'_{\mathrm{b}} \;\; \boldsymbol{a}_{\mathrm{t}}] \in \mathbf{R}^{(k+q)\times 1}$.

One can, therefore, construct a pixel reconstruction detection operator to perform subpixel target detection on an HSI, and the operator can be expressed as

$$D_{\mathrm{HSPRD}}(\boldsymbol{x}) = \exp\left[-\rho\cdot\left(\frac{\|\boldsymbol{x} - \boldsymbol{D}\boldsymbol{a}\|_1}{\|\boldsymbol{x} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}_{\mathrm{b}}\|_1}\right)^{\gamma}\right] \tag{20}$$

where ρ and γ are scale coefficients, and $\|\boldsymbol{x} - \boldsymbol{D}\boldsymbol{a}\|_1$ represents the reconstruction error when a pixel is reconstructed using endmember matrix $\boldsymbol{D}$ containing target and background feature spectra and the corresponding abundance coefficient $\boldsymbol{a}$. Correspondingly, $\|\boldsymbol{x} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}_{\mathrm{b}}\|_1$ represents the reconstruction error when a pixel is reconstructed using endmember matrix $\boldsymbol{D}_{\mathrm{b}}$ containing only the background feature spectrum and the corresponding abundance coefficient $\boldsymbol{a}_{\mathrm{b}}$.

For the convenience of expression, let

$$G(\boldsymbol{x}) = \frac{\|\boldsymbol{x} - \boldsymbol{D}\boldsymbol{a}\|_1}{\|\boldsymbol{x} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}_{\mathrm{b}}\|_1}. \tag{21}$$

The value of $G(\boldsymbol{x})$ can be discussed in two situations.

First, when pixel $\boldsymbol{x}$ is a background pixel and does not contain the target feature spectrum, the error of the reconstruction obtained by using the abovementioned two reconstruction methods should satisfy

$$\|\boldsymbol{x} - \boldsymbol{D}\boldsymbol{a}\|_1 \approx \|\boldsymbol{x} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}_{\mathrm{b}}\|_1. \tag{22}$$

We, therefore, have

$$G(\boldsymbol{x}) \approx 1. \tag{23}$$

Then, when pixel $\boldsymbol{x}$ is a target pixel and contains the target feature spectrum, the error of the reconstruction obtained by

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using the abovementioned two reconstruction methods should satisfy

$$\|\boldsymbol{x} - \boldsymbol{D}\boldsymbol{a}\|_1 \ll \|\boldsymbol{x} - \boldsymbol{D}_{\mathrm{b}}\boldsymbol{a}_{\mathrm{b}}\|_1. \tag{24}$$

Hence, we have

$$G(\boldsymbol{x}) \to 0. \tag{25}$$

This is true because when the detected pixel $\boldsymbol{x}$ contains the target feature spectrum, the reconstruction using only background endmember matrix $\boldsymbol{D}_{\mathrm{b}}$ will generate a relatively large error, whereas the reconstruction using the superset, that is, the endmember matrix $\boldsymbol{D}$ containing the target feature spectrum, is more accurate and has very little error.

According to the abovementioned analysis, it can be concluded that the value of $G(\boldsymbol{x})$ is mainly concentrated around 0 and 1. In order to analyze the influence of scale coefficients ρ and γ on the performance of the hyperspectral subpixel target detector based on pixel reconstruction (HSPRD) detector, we give a set of results for $D_{\mathrm{HSPRD}}(\boldsymbol{x})$ when ρ and γ, respectively, take different values, as shown in Figs. 1 and 2.

As shown in Fig. 1, with ρ is fixed, when the value of γ is larger, the curve is closer to the upper right corner, which means that the HSPRD detector has a better performance. In detail, with the curve closer to the upper right corner, when $G(\boldsymbol{x}) \to 0$ (target present), the value of $D_{\mathrm{HSPRD}}(\boldsymbol{x})$ is closer to 1, and when $G(\boldsymbol{x}) \approx 1$ (target absent), the value of $D_{\mathrm{HSPRD}}(\boldsymbol{x})$ is closer to 0. Thus, we should choose a relatively large γ when we use the HSPRD. In addition, with $\rho = 1$, we have $D_{\mathrm{HSPRD}}(\boldsymbol{x}) = 0.37$ when $G(\boldsymbol{x}) = 1$. However, with $\rho > 1$, the values of $D_{\mathrm{HSPRD}}(\boldsymbol{x})$ are all very close to 0. Therefore, $\rho > 1$ is a scale coefficient value condition.

Fig. 2 also shows that when the value of γ is larger, the detection effect of the HSPRD on a target pixel is better. When $\rho > 1$, the increase in ρ has little effect on the background detection effect.

In practical applications, the values of the scale coefficients should generally satisfy $\rho \in [2, 3]$ and $\rho < \gamma < 10$.

In summary, when the pixel reconstruction detection operator $D_{\mathrm{HSPRD}}(\boldsymbol{x})$ is used to calculate all the pixels in an HSI, it can suppress the background pixels, highlight the target pixels, and, finally, obtain the subpixel target detection result.

V. Experimental Results and Analysis

In order to verify the performance of HSI subpixel target detection of the algorithm proposed in this article, we conducted experiments using both the simulated and real hyperspectral data. The experimental results were compared with widely used and successful HSI subpixel target detection algorithms and evaluated for both subjective effect and objective quality. The experimental environment was MATLAB R2018a software used with a 64-bit Windows 10 operating system and an Alienware R15 laptop with an eight-core Intel Core i7-7700HQ CPU having the main frequency of 2.8 GHz and 16-GB memory.

To validate the effectiveness of the algorithm, in this section, we compare the algorithm with other subpixel target detection algorithms, including SACE, CSCR, SPSMF, finite target matched filter (FTMF) [18], hierarchically Constrained Energy Minimization (hCEM) [19], and pairwise adaptive linear matched filter (PALM) [5]. Among them, SPSMF and FTMF are subpixel detectors assuming the replacement model.

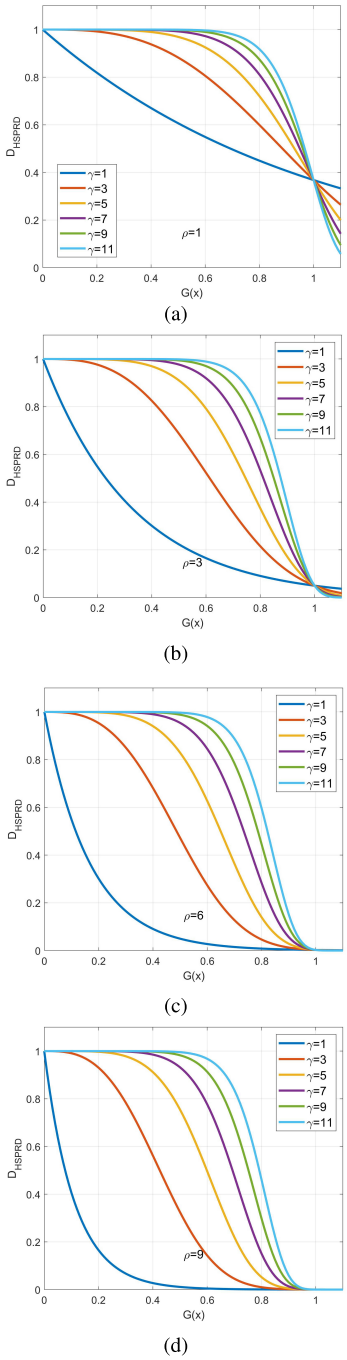


Fig. 1. Results of HSPRD detector for different values of γ when ρ is fixed. (a) $\rho = 1$. (b) $\rho = 3$. (c) $\rho = 6$. (d) $\rho = 9$.

A. Evaluation Indicators

In HSI target detection, the performance of the detectors is mainly evaluated by drawing receiver operating characteristic (ROC) curves [20]. The ROC curves indicated the change of the detection rate under different false alarm conditions where the curve closest to the upper left corner is the best. This served as an important indicator for evaluating the target detection algorithms [21].

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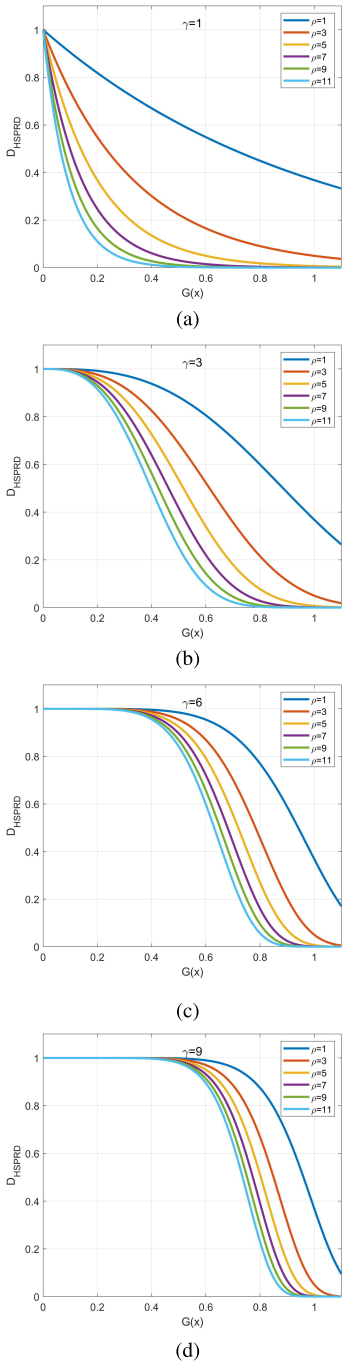


Fig. 2. Results of HSPRD detector for different values of ρ when γ is fixed. (a) $\gamma = 1$. (b) $\gamma = 3$. (c) $\gamma = 6$. (d) $\gamma = 9$.

Because an excessively high false alarm rate is meaningless in practical applications, attention is usually given to the detection performance closer to the y-axis. For this reason, the x-axis of the ROC curves in this article is in a logarithmic scale to show the performance difference of various subpixel target detection algorithms more prominently.

However, in the practical application of HSI target detection, it is sometimes difficult to establish a reliable ROC curve. The main reasons are as follows: one is that the number of pixels contained in the image scene is usually limited. The number of pixels is generally very small, resulting in limited evaluation accuracy of target detection performance.

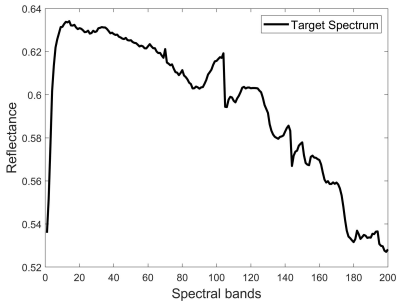


Fig. 3. Target spectrum in synthetic data.

According to the abovementioned analysis, because the ROC curve is difficult to clearly explain the advantages and disadvantages of the target detection methods in some cases, usually when the ROC curve is used to evaluate the detection performance, the area under the ROC curve (AUC) is also calculated at the same time. The AUC value can be used to evaluate the target detection method intuitively and quantitatively. The larger the value, the better the detector performance.

B. Synthetic Data Experiment

In this section, we constructed synthetic data using an HSI of the San Diego Airport in California, USA [22]. The data set was obtained by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS); the spectral band covered 224 spectral channels in the 0.4–2.5-μm range. For the ease of processing and analysis, subimages of 120 $\times$ 140 pixels were selected from the original San Diego Airport HSI data to construct the simulated data. Before artificially inserting target pixels, low signal-to-noise ratio (SNR) and water vapor absorption bands were removed from the selected hyperspectral data, leaving 200 spectral bands.

The target spectrum was a randomly selected aircraft spectral curve in the original image (see Fig. 3). Simulated data containing target points were obtained by adding different proportions of the target spectrum to 100 points in the selected image data. Specifically, ten rows were selected from the top to bottom in the selected image, and ten points were selected at equal intervals in each row for adding the target spectrum. The same proportion of the target spectrum was added in the target points on the same row, and the proportions from top to bottom were 100%, 90%, 20%, and 10%, respectively. At the same time, the proportion of the original background spectrum at the point involved was correspondingly reduced. The constructed synthetic data are shown in Fig. 4(a), while the real feature distribution is shown in Fig. 4(b). To verify the performance of the HSPRD under different noise conditions, we take the white Gaussian noise into account in the experiment, with three levels of SNR, namely, 35, 40, and 45 dB.

Fig. 5 shows the ROC curves of the detection results obtained by different detectors under different SNR conditions. As can be seen in Fig. 5, the proposed HSPRD showed distinct advantages compared with hCEM, SPSMF, SACE, and CSCR algorithms under all SNR conditions. Compared with FTMF, the performance of HSPRD is better when SNR $=$ 35 and 40 dB. With SNR $=$ 45 dB, the HSPRD has obviously better

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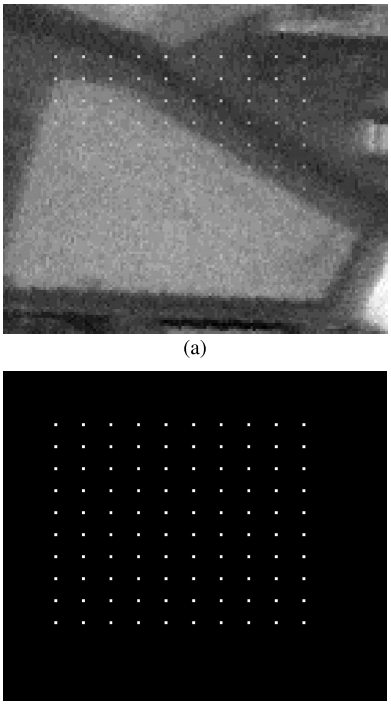


Fig. 4. (a) Synthetic hyperspectral data and (b) target distribution (target ground truth).

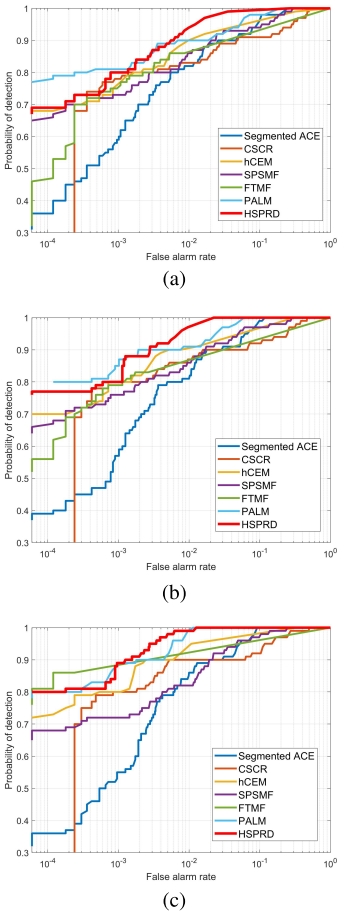


Fig. 5. Receiver operation characteristics (ROC) curves of detection results obtained by different hyperspectral subpixel target detection algorithms on the simulated data set under different SNR conditions. (a) SNR = 35 dB. (b) SNR = 40 dB. (c) SNR = 45 dB.

detection performance than FTMF when the false alarm rate is larger than 0.1%. Compared with PALM, the ROC curve of the proposed algorithm is closer to the upper left-hand corner and has clear advantages with SNR = 40 and 45 dB. In addition, when the false alarm rate is around 1%, HSPRD has the highest detection performance under all three SNR conditions. The performance of the HSPRD can be also confirmed from the AUC values of the different subpixel target detectors given in Table I. From Table I, it can be seen that the proposed HSPRD has the highest AUC values with all three SNRs, indicating that its performance is better than that of other algorithms and that it has obtained the best detection effect on the simulated data set.

Fig. 6 shows the relationship between the scale coefficients values and AUC for HSPRD under different noise conditions. From Fig. 6, when we choose a smaller ρ and a larger γ, the AUC can achieve high value. It also proves the correctness of the scale coefficients value rules discussed in Section IV.

The computational time of different subpixel detectors is shown in Table II. It can be seen that compared with the replacement model detector FTMF, the running time of HSPRD is much shorter. However, the time cost of HSPRD is significantly higher than that of SACE, SPSMF, PALM, and hCEM. Obviously, HSRPD trades for detection accuracy for time.

TABLE I
AUC VALUES OF DIFFERENT SUBPIXEL TARGET DETECTION ALGORITHMS

Method	AUC value		
	SNR = 35 dB	SNR = 40 dB	SNR = 45 dB
SACE	0.9871	0.9912	0.9922
SPSMF	0.9862	0.9897	0.99909
FTMF	0.9290	0.9147	0.9300
CSCR	0.9684	0.9759	0.9814
PALM	0.9932	0.9971	0.9992
hCEM	0.9807	0.9851	0.9908
HSPRD	**0.9969**	**0.9990**	**0.9995**

Note: SACE, segmented adaptive coherence estimator; SPSMF, subpixel spectral matched filter; FTMF, finite target matched filter; CSCR, combined sparse and collaborative representation; PALM, pairwise adaptive linear matched filter; hCEM, hierarchically Constrained Energy Minimization; HSPRD, hyperspectral subpixel target detector based on pixel reconstruction.

C. Real Data Experiment

1) RIT Data Set: In this section, the real-world data set used here is the well-studied Rochester Institute of Technology (RIT) Target Detection Test. This data set was obtained over Cooke City, MT, USA, by using the hyperspectral sensor HyMAP in 2006. It has 280 × 800 pixels with a spectral sampling interval of 15 nm and a total of 126 spectral bands covering the wavelengths of 450–2450 nm. The spatial resolution of the image is 3 m × 3 m, as shown in Fig. 7. To facilitate processing and analysis, a subimage of 100 × 100 pixels was selected from the original HSI data for subpixel target detection experiments. The selected area is shown in Fig. 8, which is the part inside the cyan box in Fig. 7.

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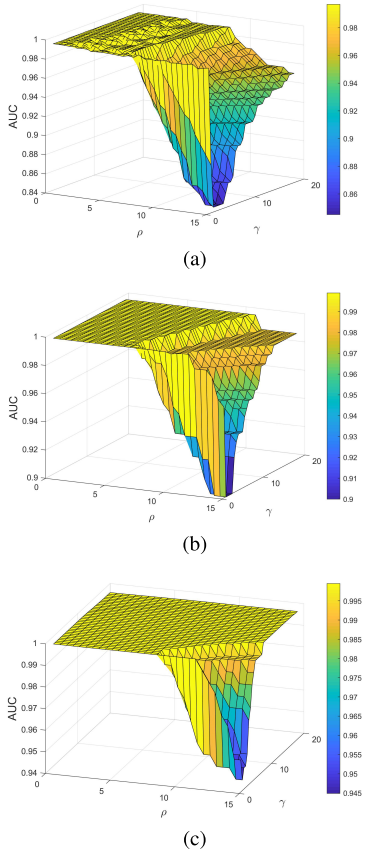


Fig. 6. AUC as functions of scale coefficients γ and ρ for HSPRD under different SNR conditions. (a) SNR = 35 dB. (b) SNR = 40 dB. (c) SNR = 45 dB.

TABLE II
COMPUTATIONAL TIME (SECONDS) OF DIFFERENT SUBPIXEL TARGET DETECTION ALGORITHMS ON THE SIMULATED DATA SET (SNR = 40)

Method	SACE	SPSMF	FTMF	CSCR	PALM	hCEM	HSPRD
Time(s)	5.728	1.184	152.870	14.593	2.201	6.976	76.685

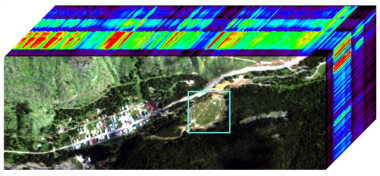


Fig. 7. RIT target detection test 3-D cube.

a) Blue Cotton: First, we select Blue Cotton as the target object. The ground truth is shown in Fig. 9(a), which includes two targets with sizes of 2 m × 2 m and 1 m × 1 m, and their indexes are named F3a and F3b, respectively. In this data set, the target feature spectrum is known [see Fig. 9(b)].

Fig. 10 shows the detection results of different subpixel detectors applied to the Blue Cotton targets. Because there are few target pixels in the scene, it is difficult for the ROC curve to accurately and reliably reflect the performance of the detectors. Therefore, the detection performance of different algorithms is only qualitatively analyzed based on the detection results shown in Fig. 10. It can be seen from the detection results that the CSCR and SPSMF algorithms



Fig. 8. Selected subimage in the RIT target detection test.

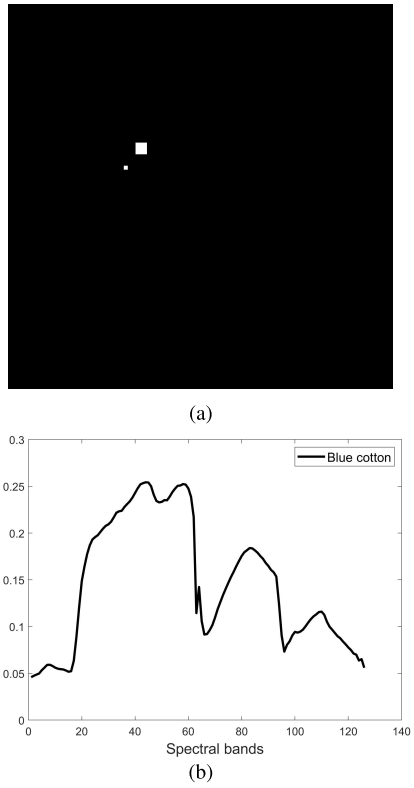


Fig. 9. (a) Ground truth and (b) target spectrum of Blue Cotton.

fail to detect both F3a and F3b. SACE, hCEM, and PALM can only detect the larger target F3a. However, both the hCEM and PALM algorithms have high false alarm detection rates. FTMF can detect the targets, but it is not enough to highlight the smaller target F3b. The proposed HSPRD algorithm can obtain very good detection results for two targets of different sizes, and it can well separate the targets from the background. Especially, for the smaller target F3b, its detection performance is significantly better than other comparison algorithms. Therefore, the proposed HSPRD can better suppress the background and highlight the target and can achieve the best detection performance for the Blue Cotton targets.

b) Red Nylon: Second, we select Red Nylon as the target object. The ground truth is shown in Fig. 11(a), which includes two targets with sizes of 2 m × 2 m and 1 m × 1 m, and their

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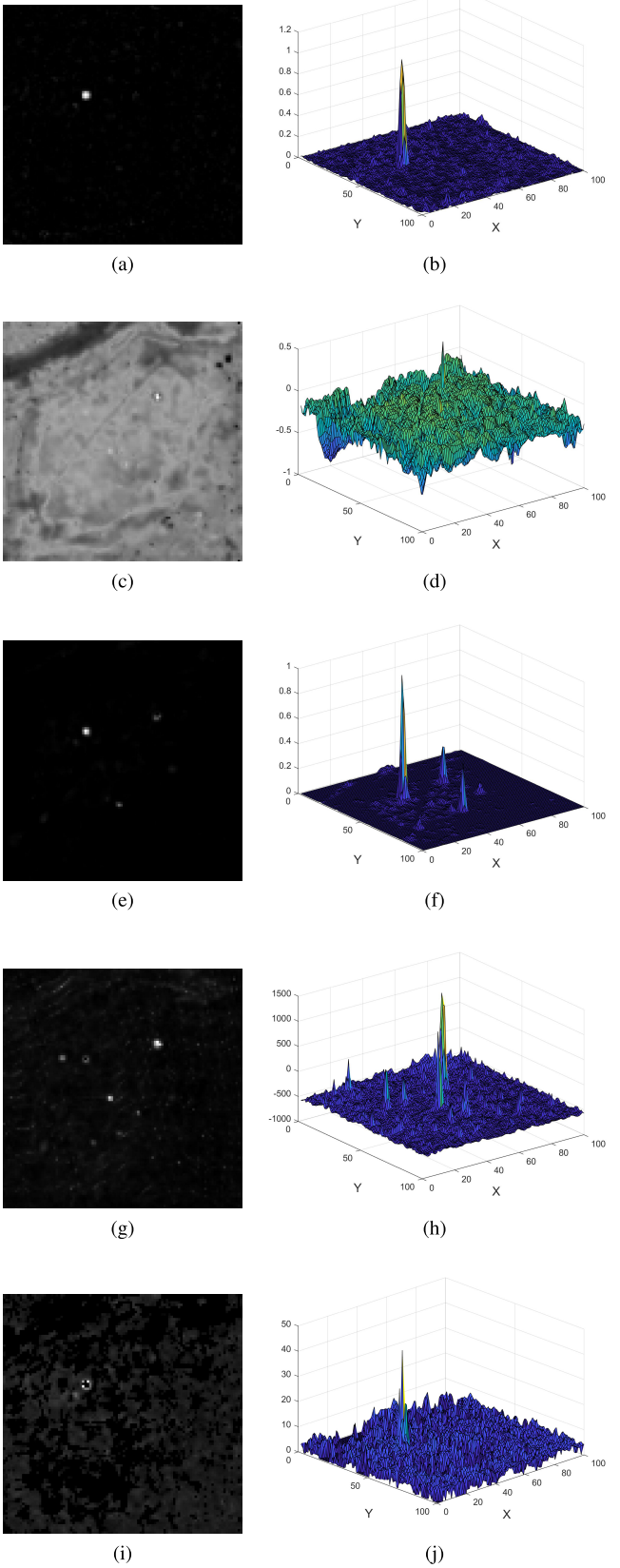


Fig. 10. Detection results of different subpixel target detection algorithms applied to Blue Cotton targets. (a) 2-D of SACE. (b) 3-D of SACE. (c) 2-D of CSCR. (d) 3-D of CSCR. (e) 2-D of hCEM. (f) 3-D of hCEM. (g) 2-D of SPSMF. (h) 3-D of SPSMF. (i) 2-D of FTMF. (j) 3-D of FTMF. (k) 2-D of PALM. (l) 3-D of PALM. (m) 2-D of HSPRD. (n) 3-D of HSPRD.

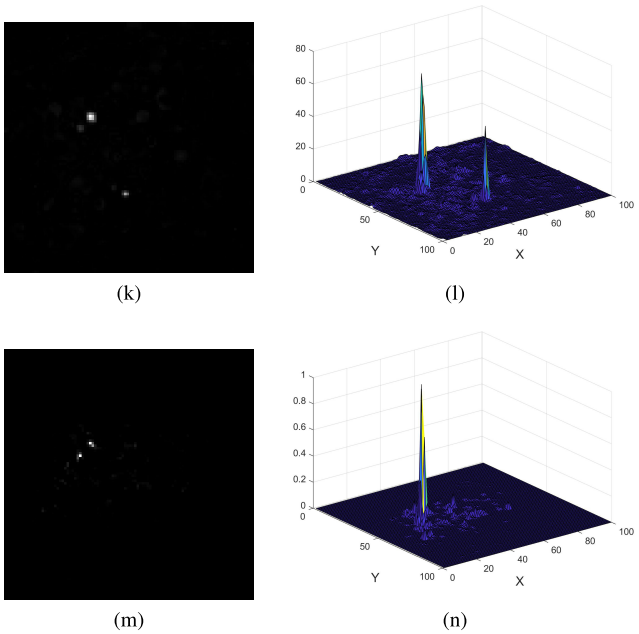


Fig. 10. *(Continued)* Detection results of different subpixel target detection algorithms applied to Blue Cotton targets. (a) 2-D of SACE. (b) 3-D of SACE. (c) 2-D of CSCR. (d) 3-D of CSCR. (e) 2-D of hCEM. (f) 3-D of hCEM. (g) 2-D of SPSMF. (h) 3-D of SPSMF. (i) 2-D of FTMF. (j) 3-D of FTMF. (k) 2-D of PALM. (l) 3-D of PALM. (m) 2-D of HSPRD. (n) 3-D of HSPRD.

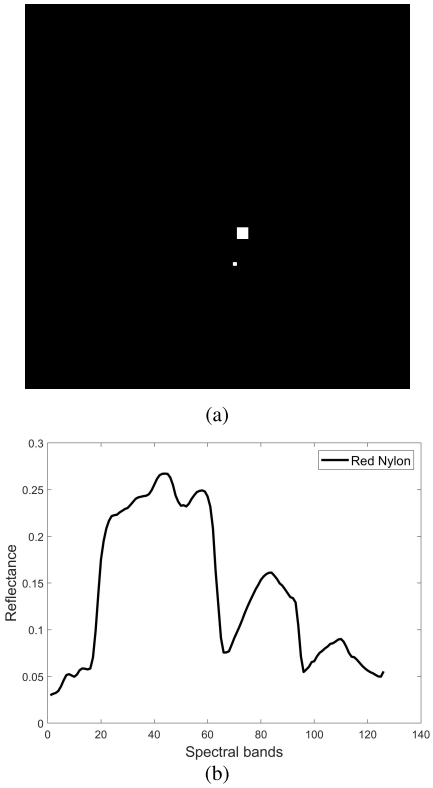


Fig. 11. (a) Ground truth and (b) target spectrum of Red Nylon.

indexes are named F4a and F4b, respectively. Also, the target feature spectrum is known [see Fig. 11(b)].

Fig. 12 shows the detection results of different subpixel target detection algorithms applied to the Red Nylon targets. As with Blue Cotton, the performance of different detectors

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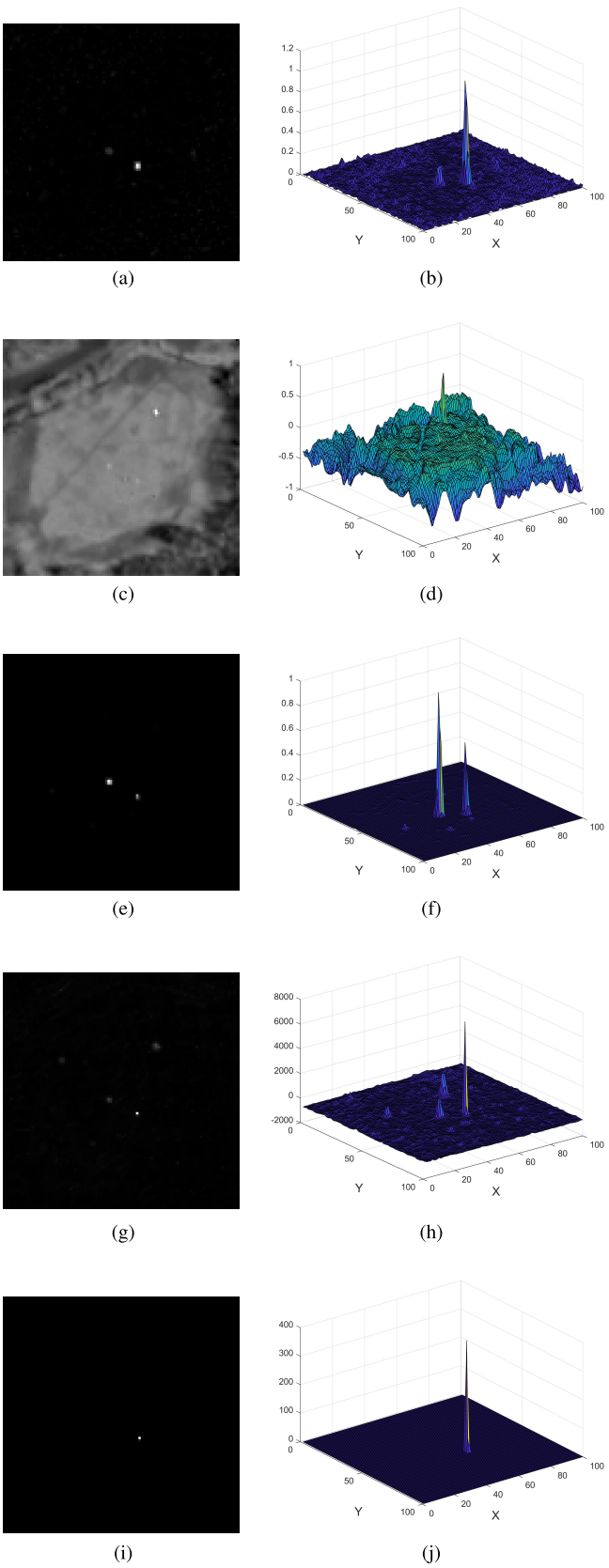


Fig. 12. Detection results of different subpixel target detection algorithms applied to Red Nylon targets. (a) 2-D of SACE. (b) 3-D of SACE. (c) 2-D of CSCR. (d) 3-D of CSCR. (e) 2-D of hCEM. (f) 3-D of hCEM. (g) 2-D of SPSMF. (h) 3-D of SPSMF. (i) 2-D of FTMF. (j) 3-D of FTMF. (k) 2-D of PALM. (l) 3-D of PALM. (m) 2-D of HSPRD. (n) 3-D of HSPRD.

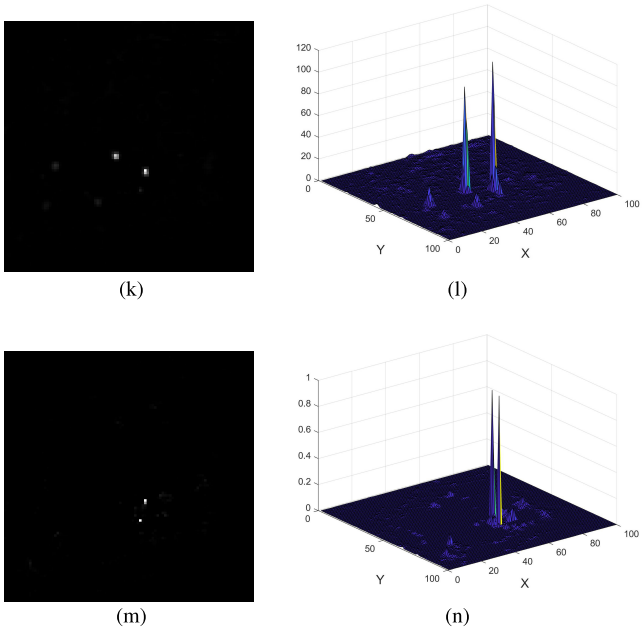


Fig. 12. *(Continued)* Detection results of different subpixel target detection algorithms applied to Red Nylon targets. (a) 2-D of SACE. (b) 3-D of SACE. (c) 2-D of CSCR. (d) 3-D of CSCR. (e) 2-D of hCEM. (f) 3-D of hCEM. (g) 2-D of SPSMF. (h) 3-D of SPSMF. (i) 2-D of FTMF. (j) 3-D of FTMF. (k) 2-D of PALM. (l) 3-D of PALM. (m) 2-D of HSPRD. (n) 3-D of HSPRD.

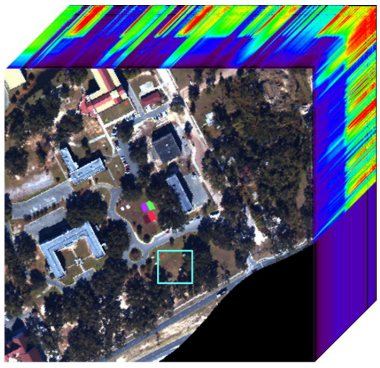


Fig. 13. MUUFL Gulfport data set 3-D cube.

is only qualitatively analyzed based on the detection results shown in Fig. 12. It can be seen from the detection results that the CSCR algorithm fails to detect both F4a and F4b. SACE, hCEM, and PALM can only detect the larger target F4a but fail to detect the small-sized target F4b because of the same false target. SPSMF and FTMF can also only detect the larger target F4a but no false targets. HSPRD can obtain very good detection results for both two targets of different sizes, and it can well separate the targets from the background. Thus, the proposed HSPRD can achieve the best performance for the Red Nylon targets.

2) MUUFL Gulfport Data Set: In this section, the MUUFL Gulfport data set [23] is used for real data experiments. This data set was acquired over the Gulf Park Campus of the University of Southern Mississippi using airborne hyperspectral sensors. It has 325 × 337 pixels with a spectral sampling interval of 9.5 nm and a total of 72 spectral bands covering the wavelengths of 367.7–1043.4 nm. The spatial resolution

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Fig. 14. Selected subimage in the MUUFL Gulfport data set.

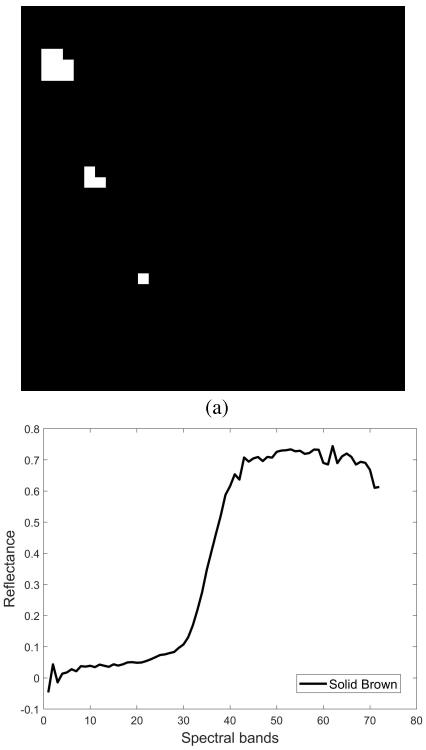


Fig. 15. (a) Ground truth and (b) target spectrum of Solid Brown.

of the image is 1 m × 1 m, as shown in Fig. 13. Throughout the scene, four different colors of cotton fabrics, such as Solid Dark Green, Vineyard Green, Solid Brown, and Pea Green, were used for man-made target objects. The target size includes three specifications of 3 m × 3 m, 1 m × 1 m, and 0.5 m × 0.5 m. A total of 60 human-made targets are set in the scene, which leads to the more challenging subpixel target detection problem. Among them, the small size target of 0.5 m × 0.5 m is used to simulate the target object with less than the spatial resolution of the hyperspectral sensor. The medium size target of 1 m × 1 m is the same as the spatial resolution and can theoretically constitute a pure target pixel. However, this is almost impossible in practical situations; usually, the target will exist in several adjacent pixels with different target spectral abundances. A large size target of 3 m × 3 m is used to ensure that at least a pure target

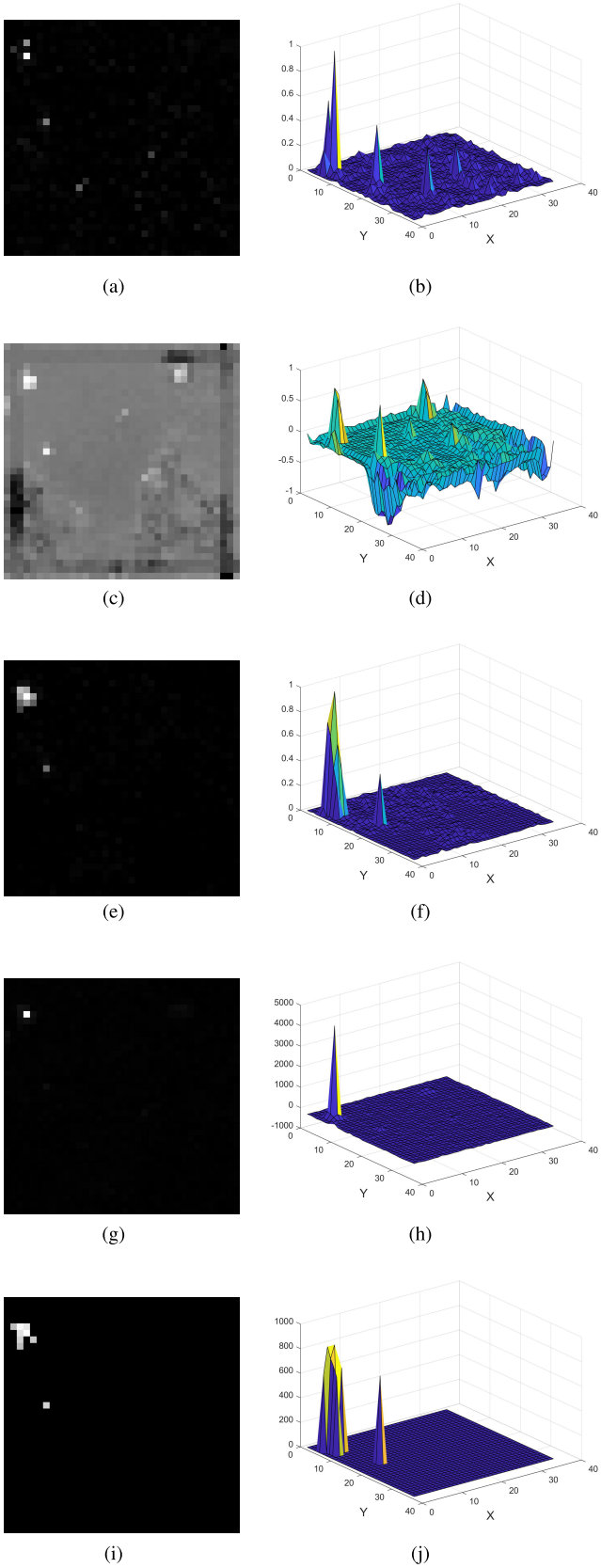


Fig. 16. Detection results of different subpixel target detection algorithms applied to the MUUFL Gulfport data set. (a) 2-D of SACE. (b) 3-D of SACE. (c) 2-D of CSCR. (d) 3-D of CSCR. (e) 2-D of hCEM. (f) 3-D of hCEM. (g) 2-D of SPSMF. (h) 3-D of SPSMF. (i) 2-D of FTMF. (j) 3-D of FTMF. (k) 2-D of PALM. (l) 3-D of PALM. (m) 2-D of HSPRD. (n) 3-D of HSPRD.

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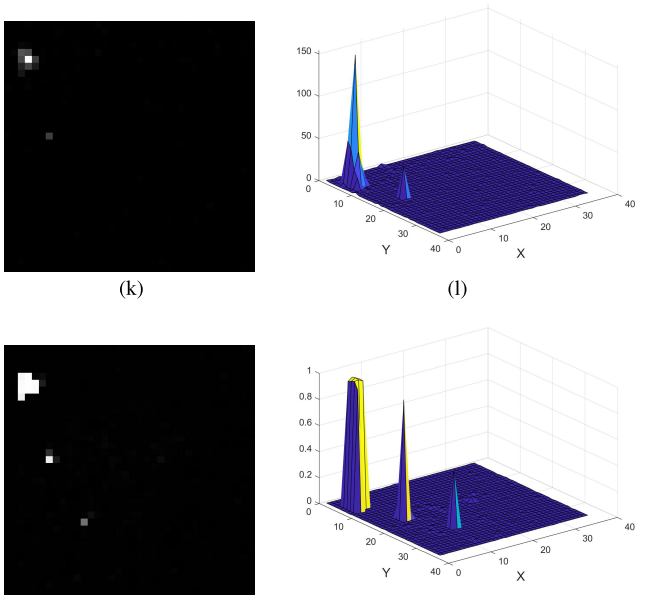


Fig. 16. *(Continued)* Detection results of different subpixel target detection algorithms applied to the MUUFL Gulfport data set. (a) 2-D of SACE. (b) 3-D of SACE. (c) 2-D of CSCR. (d) 3-D of CSCR. (e) 2-D of hCEM. (f) 3-D of hCEM. (g) 2-D of SPSMF. (h) 3-D of SPSMF. (i) 2-D of FTMF. (j) 3-D of FTMF. (k) 2-D of PALM. (l) 3-D of PALM. (m) 2-D of HSPRD. (n) 3-D of HSPRD.

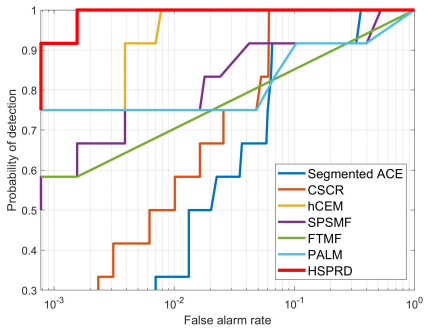


Fig. 17. ROC curves of detection results obtained by different hyperspectral subpixel target detection algorithms on the MUUFL Gulfport data set.

pixel exists. To facilitate processing and analysis, a subimage of 36 × 36 pixels was selected from the original HSI data for subpixel target detection experiments. The selected area is shown in Fig. 14, which is the part inside the cyan box in Fig. 13.

In the experiment, we select Solid Brown as the target object. The real feature distribution is shown in Fig. 15(a), and the target feature spectrum is known [see Fig. 15(b)].

Fig. 16 shows the detection results of different subpixel target detection algorithms applied to the Solid Brown targets. It can be seen from the detection results that due to the existence of pure pixels, all the target detectors have good detection results for large-scale targets in the upper left corner, and the existence of targets can be found. For the middle size target in the middle of the scene, without pure pixels, the detection performance of SPSMF and PALM is not good, but, because the target object size is still large, its proportion in mixed pixels is high, and other remaining algorithms can obtain satisfactory detection performance. However, for the small-scale target located at the bottom of the scene, because the target object only accounts for one-quarter of a pixel and even the existence of the target object cannot be found by using the color information in the image, it needs detector to be able to find the weak target signal in mixed pixels. According to the detection results, SPSMF, FTMF, PALM, hCEM, and PALM algorithm cannot detect the small-scale target. SACE can detect the target, but it can also detect multiple false targets at the same time, and the detection accuracy is not high. The proposed HSPRD can obtain very good detection results for three different size targets and can separate the targets from the background very well, especially for small subpixel targets. The detection performance of HSPRD is obviously better than other comparison algorithms. Therefore, the proposed HSPRD can achieve the best detection effect for the Solid Brown targets.

TABLE III

AUC VALUES OF DIFFERENT SUBPIXEL TARGET DETECTION ALGORITHMS ON THE MUUFL GULFPORT DATA SET

Method	SACE	SPSMF	FTMF	CSCR	PALM	hCEM	HSPRD
AUC value	0.9483	0.9568	0.7913	0.9798	0.9299	0.9987	0.9997

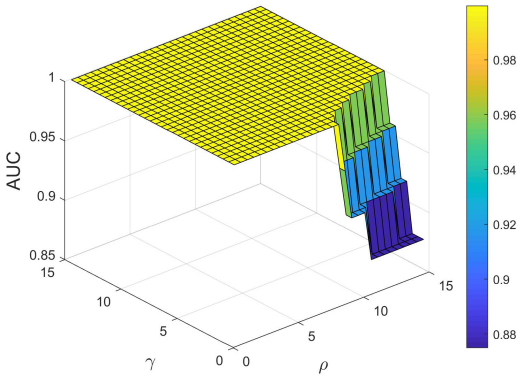


Fig. 18. AUC as functions of scale coefficients γ and ρ for HSPRD on the MUUFL Gulfport data set.

To quantitatively evaluate the detection performance of the HSPRD algorithm, we show the ROC curves of the detection results obtained by different hyperspectral subpixel target detection algorithms on the simulated data set (see Fig. 17). Fig. 17 shows that compared with the SPSMF, FTMF, SACE, hCEM, PALM, and CSCR algorithms, the proposed HSPRD algorithm has obvious advantages. It is worth noting that the false alarm rate of SACE is high, which also indicates that although it can detect all target features, the detection accuracy is low. The AUC values of the different subpixel target detection algorithms also confirm the advantages of HSPRD over other algorithms from another perspective (see Table III). This table shows that the proposed algorithm has the highest AUC value, indicating that it performed better than the other competing algorithms and that it can obtain the optimal detection results on a real data set.

Fig. 18 shows the relationship between the scale coefficients values and AUC for HSPRD on the MUUFL Gulfport data set. It can be seen that when we follow the rules (discussed in Section IV) to choose coefficient values, the AUC can achieve a good value.

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VI. Conclusion

In this article, we proposed a subpixel target detection method based on background endmember extraction of hyperspectral remote sensing images. First, we extracted endmembers from HSIs to be tested using an endmember extraction algorithm based on robust nonnegative dictionary learning. We calculated the corresponding abundance coefficient and introduced a detection operator based on pixel reconstruction. Pixel-by-pixel target detection was carried out on the HSIs to be detected using the previously acquired background endmember and the known target feature spectrum, and finally, the detection result was obtained. Compared with the current state-of-the-art subpixel target detection algorithm for HSIs, the proposed method has better detection performance on both the simulated data set and the real data set. However, because the proposed algorithm uses dictionary learning, it has a greater time overhead than conventional subpixel target detection algorithms. The next step of the research will focus on the rapid implementation of the algorithm.

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